Кочеткова М. П. Интегралы/часть3

Часть 3.

8.3.19

$$\int \frac{5dx}{x+\sqrt{2}} = 5\int \frac{dx}{x+\sqrt{2}} = \left[d(x+\sqrt{2}) = dx\right] = 5\cdot\int \frac{d(x+\sqrt{2})}{x+\sqrt{2}} =$$

$$= \left[\int \frac{dx}{x} = \ln|x| + C\right] = 5\cdot\ln|x+\sqrt{2}| + C.$$

Интеграл I типа, т.к 1) x - в 1 степени
2) $x-a$ - в 1 степени
$x-a = x+\sqrt{2} \Rightarrow a = -\sqrt{2}$

8.3.20

$$\int \frac{4dx}{(x-\frac{1}{2})^3} = 4\cdot\int \frac{dx}{(x-\frac{1}{2})^3} = \left[d(x-\tfrac{1}{2}) = dx\right] = 4\cdot\int \frac{d(x-\frac{1}{2})}{(x-\frac{1}{2})^3} =$$

$$= \left[\int x^{\alpha}dx = \frac{x^{\alpha+1}}{\alpha+1} + C\right] = \frac{4(x-\frac{1}{2})^{-3+1}}{-3+1} + C = -\frac{4}{2(x-\frac{1}{2})^2} + C =$$

$$= -\frac{2}{(x-\frac{1}{2})^2} + C.$$

Интеграл II типа, т.к 1) x - в 1 степени
2) $x-a = x-\frac{1}{2}$ - во 3 степени
$a = \frac{1}{2}$

8.3.21

$$\int \frac{7dx}{(x+3)^6} = 7\cdot\int \frac{dx}{(x+3)^6} = \left[d(x+3) = dx\right] = 7\cdot\int \frac{d(x+3)}{(x+3)^6} =$$

$$= \left[\int x^{\alpha}dx = \frac{x^{\alpha+1}}{\alpha+1} + C\right] = \frac{7\cdot(x+3)^{-6+1}}{-6+1} + C = -\frac{7}{5(x+3)^5} + C$$

Интеграл II типа, т.к. 1) x - в 1 степени
2) $x-a = x+3$ - в 6 степени
$a = -3$

8.3.22

$$\int \frac{dx}{(3x+2)^4} = \left[\begin{aligned} t &= 3x+2 \Rightarrow dt = d(3x+2) = \\ &= (3x+2)'_x\, dx = 3dx \Rightarrow dx = \frac{dt}{3} \end{aligned} \right] = \int \frac{dt}{3t^4} = \frac{1}{3} \int \frac{dt}{t^4} =$$

$$= \left[\int x^\alpha dx = \frac{x^{\alpha+1}}{\alpha+1} + C \right] = \frac{1}{3} \cdot \frac{t^{-4+1}}{-4+1} + C = -\frac{1}{9t^3} + C =$$

$$= -\frac{1}{9(3x+2)^3} + C.$$

Интеграл II типа, т.к. 1) x - в 1 степени
2) $x - a = 3x+2$ - в 4 степени
$x - a = x + \frac{2}{3} \Rightarrow a = -\frac{2}{3}$

8.3.23

$$\int \frac{dx}{x^2-4x+8} = \left[\begin{aligned} & x^2-4x+8=0 \\ & D = 16-32 = -16 < 0 \Rightarrow \text{нет корней} \end{aligned} \right] \Rightarrow \text{Интеграл } \mathrm{III} \text{ типа}$$

$[\; A \cdot x + B = 0 \cdot x + 1$

$A = 0, \quad B = 1, \quad p = -4, \quad q = 8$

$(x^2-4x+8)'_x = 2x - 4$ - нет в числ. $\Rightarrow$ нах. реш. 2-го инт.

$y = x + p/2 = x - 4/2 = x - 2 \Rightarrow dy = dx$

$a = \sqrt{q - p^2/4} = \sqrt{8 - \frac{16}{4}} = \sqrt{8-4} = \sqrt{4} = 2$

тогда

$x^2 - 4x + 8 = y^2 + 2^2 \;] =$

$$= \int \frac{dy}{y^2+2^2} = \left[\int \frac{dx}{x^2+a^2} = \frac{1}{a} \cdot \operatorname{arctg} \frac{x}{a} + C \right] =$$

$$= \frac{1}{2} \cdot \operatorname{arctg} \frac{y}{2} + C = \frac{1}{2} \cdot \operatorname{arctg} \frac{x-2}{2} + C$$

8.3.24

$$\int \frac{dx}{x^2+x+1} = \left[\begin{array}{l} x^2+x+1=0 \\ D=1-4=-3<0 \Rightarrow \text{нет корней} \end{array}\right] \Rightarrow \text{интеграл III типа}$$

$[\, A\cdot x + B = 0\cdot x + 1$

$A=0 \quad , \quad B=1 \quad , \quad p=1 \quad , \quad q=1$

$(x^2+x+1)'_x = 2x+1$ - нет в числ. $\Rightarrow$ нах реш 2-го инт.

$y = x + p/2 = x + \frac{1}{2} \Rightarrow dy = dx$

$a = \sqrt{q - p^2/4} = \sqrt{1-\frac{1}{4}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$

тогда

$$x^2+x+1 = y^2 + \left(\frac{\sqrt{3}}{2}\right)^2 \quad] =$$

$$= \int \frac{dy}{y^2+\left(\frac{\sqrt{3}}{2}\right)^2} = \left[\int \frac{dx}{x^2+a^2} = \frac{1}{a}\operatorname{arctg}\frac{x}{a} + C \right] =$$

$$= \frac{2}{\sqrt{3}} \operatorname{arctg} \frac{2y}{\sqrt{3}} + C = \frac{2}{\sqrt{3}} \operatorname{arctg} \frac{2(x+\frac{1}{2})}{\sqrt{3}} + C =$$

$$= \frac{2\sqrt{3}}{3} \operatorname{arctg} \frac{(2x+1)\sqrt{3}}{3} + C$$

8.3.25

$$\int \frac{6x+1}{x^2-8x+25}\,dx = \left[\begin{array}{l} x^2-8x+25=0 \\ D=64-100=-36<0 \Rightarrow \text{нет корней} \end{array}\right] \Rightarrow \text{интеграл III типа}$$

$[\, A=6 \quad , \quad B=1 \quad , \quad p=-8 \quad , \quad q=25$

$Ax+B = \frac{6}{2}\cdot(2x-8) + 1 - \frac{6\cdot(-8)}{2}$

$(x^2-8x+25)'_x = 2x-8$ - есть в числ $\Rightarrow$ нах реш 1-го инт.

$t = x^2-8x+25 \Rightarrow dt = (2x-8)\,dx =$

$$\int \frac{(2x-8)\,dx}{x^2-8x+25} = \int \frac{dt}{t} = \ln|t| + C = \ln(x^2-8x+25) + C$$

реш 2-го инт.

$$y = x + p/2 = x - 8/2 = x - 4 \Rightarrow dy = dx$$

$$a = \sqrt{q - \frac{p^2}{4}} = \sqrt{25 - \frac{64}{4}} = \sqrt{25-16} = \sqrt{9} = 3$$

тогда

$$x^2 + x + 1 = y^2 + 3^2 \ \big] =$$

$$= \frac{6}{2} \cdot \ln(x^2-8x+25) + \left(1 - \frac{6\cdot(-8)}{2}\right) \cdot \frac{1}{3} \operatorname{arctg} \frac{y}{3} + C =$$

$$= 3 \cdot \ln(x^2-8x+25) + \frac{25}{3} \operatorname{arctg} \frac{x-4}{3} + C$$

8.3.26

$$\int \frac{5x+2}{x^2+2x+10}\,dx = \left[\begin{array}{l} x^2+2x+10=0 \\ D = 4-40 = -36 < 0 \Rightarrow \text{нет корней} \end{array}\right] \Rightarrow \text{интеграл III типа}$$

$[\ A = 5, \quad B = 2, \quad p = 2, \quad q = 10$

$$Ax + B = \frac{5}{2} \cdot (2x+2) + 2 - \frac{5\cdot 2}{2}$$

$(x^2+2x+10)'_x = 2x+2$ - есть в числ $\Rightarrow$ нах реш. 1-го инт.

$$t = x^2+2x+10 \Rightarrow dt = (2x+2)\,dx \Rightarrow$$

$$\int \frac{(2x+2)\,dx}{x^2+2x+10} = \int \frac{dt}{t} = \ln|t| + C = \ln(x^2+2x+10) + C$$

реш 2-го инт

$$y = x + p/2 = x + 2/2 = x + 1 \Rightarrow dy = dx$$

$$a = \sqrt{q - \frac{p^2}{4}} = \sqrt{10 - \frac{2^2}{4}} = \sqrt{10-1} = \sqrt{9} = 3$$

тогда

$$x^2+2x+10 = y^2+3^2 \ \Big] =$$

$$= \frac{5}{2}\cdot \ln(x^2+2x+10) + \left(2 - \frac{5\cdot 2}{2}\right)\cdot \frac{1}{3} \operatorname{arctg} \frac{y}{3} + C =$$

$$= \frac{5}{2}\cdot \ln(x^2+2x+10) + (2-5)\cdot \frac{1}{3} \operatorname{arctg} \frac{x+1}{3} + C =$$

$$= \frac{5}{2}\cdot \ln(x^2+2x+10) - \operatorname{arctg} \frac{x+1}{3} + C$$

8.5.27

$$\int \frac{x+2}{x^2+3x+5}\,dx = \left[\begin{array}{l} x^2+3x+5=0 \\ D = 9-20 = -11 < 0 \Rightarrow \text{нет корней} \end{array}\right] \Rightarrow \text{интеграл III типа}$$

$$\Big[A=1, \quad B=2, \quad p=3, \quad q=5$$

$$Ax+B = \frac{1}{2}\cdot(2x+3) + 2 - \frac{3}{2}.$$

$(x^2+3x+5)'_x = 2x+3$ – есть в числ. $\Rightarrow$ нах реш 1-го инт.

$$t = x^2+3x+5 \Rightarrow dt = (2x+3)\,dx$$

$$\int \frac{(2x+3)dx}{x^2+3x+5} = \int \frac{dt}{t} = \ln|t| + C = \ln(x^2+3x+5) + C$$

реш. 2-го инт.

$$y = x + p/2 = x + 3/2 \Rightarrow dy = dx$$

$$a = \sqrt{q - \frac{p^2}{4}} = \sqrt{5 - \frac{9}{4}} = \sqrt{\frac{11}{4}} = \frac{\sqrt{11}}{2}$$

тогда

$$x^2+3x+5 = y^2 + \left(\frac{\sqrt{11}}{2}\right)^2 \ \Big] =$$

$$= \frac{1}{2}\ln(x^2+3x+5) + \left(2 - \frac{3}{2}\right)\cdot\frac{2}{\sqrt{11}}\,\mathrm{arctg}\,\frac{2y}{\sqrt{11}} + C =$$

$$= \frac{1}{2}\cdot\ln(x^2+3x+5) + \frac{\sqrt{11}}{11}\,\mathrm{arctg}\,\frac{2\cdot\sqrt{11}\cdot(x+3/2)}{11} + C =$$

$$= \frac{1}{2}\cdot\ln(x^2+3x+5) + \frac{\sqrt{11}}{11}\,\mathrm{arctg}\,\frac{(2x+3)\sqrt{11}}{11} + C$$

8.3.28

$$\int\frac{2x-1}{5x^2+2x+1}\,dx = \frac{1}{5}\cdot\int\frac{2x-1}{x^2+\frac{2}{5}x+\frac{1}{5}}\,dx = \left[\begin{array}{l} x^2+\frac{2}{5}x+\frac{1}{5}=0 \\ D=\frac{4}{25}-\frac{4}{5}=-\frac{16}{25}<0 \Rightarrow \text{нет корней}\end{array}\right] \Rightarrow$$

$\Rightarrow$ интеграл $\underline{III}$ типа

[$A=2$, $B=-1$, $p=\frac{2}{5}$, $q=\frac{1}{5}$

$$Ax+B = \frac{2}{2}\cdot\left(2x+\frac{2}{5}\right) - 1 - \frac{2\cdot 2}{2\cdot 5} = \left(2x+\frac{2}{5}\right) - \frac{7}{5}$$

$\left(x^2+\frac{2}{5}x+\frac{1}{5}\right)'_x = 2x+\frac{2}{5}$ - есть в числ. $\Rightarrow$ нах. реш. 1-го инт.

$$t = x^2+\frac{2}{5}x+\frac{1}{5} \Rightarrow dt = \left(2x+\frac{2}{5}\right)dx$$

$$\int\frac{\left(2x+\frac{2}{5}\right)}{x^2+\frac{2}{5}x+\frac{1}{5}} = \int\frac{dt}{t} = \ln|t| + C = \ln\left(x^2+\frac{2}{5}x+\frac{1}{5}\right) + C$$

реш. 2-го инт.

$$y = x + p/2 = x + \frac{2}{5\cdot 2} = x + 1/5 \Rightarrow dy = dx$$

$$a = \sqrt{q - \frac{p^2}{4}} = \sqrt{\frac{1}{5} - \frac{4}{25\cdot 4}} = \sqrt{\frac{20}{100} - \frac{4}{100}} = \sqrt{\frac{16}{100}} = \frac{4}{10} = \frac{2}{5}$$

тогда

$$x^2+\frac{2}{5}x+\frac{1}{5} = y^2 + \left(\frac{2}{5}\right)^2 \ ] =$$

$$= \frac{1}{5}\left(\frac{2}{2}\cdot\ln\left(x^2+\frac{2}{5}x+\frac{1}{5}\right) + \left(-1-\frac{2\cdot 2}{2\cdot 5}\right)\cdot\frac{1\cdot 5}{2}\,\mathrm{arctg}\,\frac{5y}{2}\right) + C =$$

$$= \frac{1}{5}\ln\left(x^2+\frac{2}{5}x+\frac{1}{5}\right) - \frac{7}{5\cdot 2}\,\mathrm{arctg}\,\frac{5\cdot\left(x+\frac{1}{5}\right)}{2} + C =$$

$$= \frac{1}{5}\ln\left(x^2+\frac{2}{5}x+\frac{1}{5}\right) - \frac{7}{10}\,\mathrm{arctg}\,\frac{5x+1}{2} + C.$$

8.5.29

$$\int \frac{x-1}{(x^2+2x+3)^2}\,dx = \left[\begin{array}{l} x^2+2x+3=0 \\ D=4-12=-8<0 \Rightarrow \text{нет корней} \\ x^2+px+q = x^2+2x+3 \ \text{- во 2-ой степени} \end{array}\right] \Rightarrow$$

$\Rightarrow$ Интеграл IV типа.

$[\ A=1,\quad B=-1,\quad p=2,\quad q=3$

$$Ax+B = \frac{1}{2}\cdot(2x+2) + \left(-1-\frac{1\cdot 2}{2}\right)$$

$(x^2+2x+3)'_x = 2x+2$ - есть в числ $\Rightarrow$ как реш 1-го инт

$$t = x^2+2x+3 \Rightarrow dt = (2x+2)dx$$

$$\int \frac{(2x+2)dx}{(x^2+2x+3)^2} = \int \frac{dt}{t^2} = \left[\int x^{\alpha}dx = \frac{x^{\alpha+1}}{\alpha+1}+C\right] = \frac{t^{-1}}{-1}+C = -\frac{1}{t}+C =$$

$$= -\frac{1}{x^2+2x+3} + C$$

реш. 2-го инт.

$$y = x+p/2 = x+2/2 = x+1 \Rightarrow dy = dx$$

$$a = \sqrt{q-\frac{p^2}{4}} = \sqrt{3-\frac{4}{4}} = \sqrt{3-1} = \sqrt{2}$$

тогда

$$(x^2+2x+3)^2 = \left(y^2+(\sqrt{2})^2\right)^2 = (y^2+2)^2$$

$$\int \frac{dy}{(y^2+2)^2} = \frac{1}{2(2-1)\cdot 2} \cdot \frac{y}{(y^2+2)} + \frac{1}{2} \cdot \frac{2\cdot 2-3}{2\cdot 2-2} \cdot \int \frac{dy}{(y^2+2)} =$$

$$= \frac{1}{4} \cdot \frac{y}{y^2+2} + \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \operatorname{arctg} \frac{y}{\sqrt{2}} + C =$$

$$= \frac{y}{4(y^2+2)} + \frac{1}{4\sqrt{2}} \operatorname{arctg} \frac{y}{\sqrt{2}} + C =$$

$$= \frac{y}{4(y^2+2)} + \frac{\sqrt{2}}{8} \operatorname{arctg} \frac{y\sqrt{2}}{2} + C$$

$$\int \frac{Ax+B}{(x^2+px+q)^n} dx = \frac{A}{2} \cdot \int \frac{dt}{t^n} + \left(B - \frac{Ap}{2}\right) \cdot \int \frac{dy}{(y^2+a^2)^n} \Big] =$$

$$= \frac{1}{2} \cdot \left(-\frac{1}{x^2+2x+3}\right) - 2 \cdot \left(\frac{y}{4(y^2+2)} + \frac{\sqrt{2}}{8} \operatorname{arctg} \frac{y\sqrt{2}}{2}\right) + C =$$

$$= -\frac{1}{2(x^2+2x+3)} - \frac{y}{2(y^2+2)} - \frac{\sqrt{2}}{4} \operatorname{arctg} \frac{y\sqrt{2}}{2} + C =$$

$$= -\frac{1}{2(x^2+2x+3)} - \frac{x+1}{2((x+1)^2+2)} - \frac{\sqrt{2}}{4} \operatorname{arctg} \frac{\sqrt{2}(x+1)}{2} + C =$$

$$= -\frac{1}{2(x^2+2x+3)} - \frac{x+1}{2(x^2+2x+3)} - \frac{\sqrt{2}}{4} \operatorname{arctg} \frac{\sqrt{2}(x+1)}{2} + C =$$

$$= -\frac{x+2}{2(x^2+2x+3)} - \frac{\sqrt{2}}{4} \operatorname{arctg} \frac{\sqrt{2}(x+1)}{2} + C$$

8.3.30

$$\int \frac{2x+1}{(x^2+2x+5)^2} dx = \left[\begin{array}{l} x^2+2x+5=0 \\ D = 4-20 = -16 < 0 \Rightarrow \text{нет корней} \\ x^2+px+q = x^2+2x+5 \text{ - во 2ой степени} \end{array}\right] \Rightarrow \text{интеграл } \overline{\text{IV}} \text{ типа}$$

[$A=2$, $B=1$, $p=2$, $q=5$

$$Ax+B=\frac{2}{2}\cdot(2x+2)+\left(1-\frac{2\cdot 2}{2}\right)$$

$(x^2+2x+5)'_x = 2x+2$ – есть в числ. => нах реш 1-го инт.

$$t=x^2+2x+5 \Rightarrow dt=(2x+2)dx$$

$$\int\frac{(2x+2)dx}{(x^2+2x+5)^2}=\int\frac{dt}{t^2}=\left[\int x^\alpha dx=\frac{x^{\alpha+1}}{\alpha+1}+C\right]=\frac{t^{-1}}{-1}+C=-\frac{1}{t}+C=$$

$$=-\frac{1}{x^2+2x+5}+C$$

реш. 2-го инт.

$$y=x+p/2=x+2/2=x+1 \Rightarrow dy=dx$$

$$a=\sqrt{q-\frac{p^2}{4}}=\sqrt{5-\frac{4}{4}}=\sqrt{4}=2$$

тогда

$$(x^2+2x+5)^2=(y^2+2^2)^2$$

$$\int\frac{dy}{(y^2+a^2)^n}=\frac{1}{2(n-1)a^2}\cdot\frac{y}{(y^2+a^2)^{n-1}}+\frac{1}{a^2}\cdot\frac{2n-3}{2n-2}\cdot\int\frac{dy}{(y^2+a^2)^{n-1}}$$

$$\int\frac{dy}{(y^2+2^2)^2}=\frac{1}{2(2-1)2^2}\cdot\frac{y}{(y^2+2^2)^{2-1}}+\frac{1}{2^2}\cdot\frac{2\cdot 2-3}{2\cdot 2-2}\cdot\int\frac{dy}{(y^2+2^2)^{2-1}}=$$

$$=\frac{1}{8}\cdot\frac{y}{y^2+4}+\frac{1}{8}\cdot\int\frac{dy}{y^2+2^2}=\left[\int\frac{dx}{x^2+a^2}=\frac{1}{a}\operatorname{arctg}\frac{x}{a}+C\right]=$$

$$=\frac{y}{8(y^2+4)}+\frac{1}{8}\cdot\frac{1}{2}\operatorname{arctg}\frac{y}{2}+C=$$

$$=\frac{x+1}{8((x+1)^2+4)}+\frac{1}{16}\operatorname{arctg}\frac{x+1}{2}+C=$$

$$= \frac{x+1}{8(x^2+2x+5)} + \frac{1}{16} \operatorname{arctg} \frac{x+1}{2} + C.$$

$$\int \frac{Ax+B}{(x^2+px+q)^n} dx = \frac{A}{2} \cdot \int \frac{dt}{t^n} + \left(B - \frac{Ap}{2}\right) \cdot \int \frac{dy}{(y^2+a^2)^n} \Bigg] =$$

$$= \frac{2}{2} \cdot \left(-\frac{1}{x^2+2x+5}\right) - 1 \cdot \left(\frac{x+1}{8(x^2+2x+5)} + \frac{1}{16} \operatorname{arctg} \frac{x+1}{2}\right) + C =$$

$$= -\frac{8}{8(x^2+2x+5)} - \frac{x+1}{8(x^2+2x+5)} - \frac{1}{16} \operatorname{arctg} \frac{x+1}{2} + C =$$

$$= -\frac{x+9}{8(x^2+2x+5)} - \frac{1}{16} \operatorname{arctg} \frac{x+1}{2} + C$$

8.3.31

$$\int \frac{dx}{(x^2+1)^4} = \left[\begin{array}{l} x^2+1=0 \\ D = 0-4 = -4 < 0 \Rightarrow \text{нет корней} \\ x^2+px+q = x^2+1 \ \text{ - в 4 степени} \end{array}\right] \Rightarrow \text{интеграл IV типа.}$$

$[\ A=0,\quad B=1,\quad p=0,\quad q=1$

$Ax+B = 0 \cdot x + 1$

$(x^2+1)'_x = 2x$ - нет в числ. $\Rightarrow$ нах реш. 2-го инт.

$y = x + p/2 = x + 0 = x \Rightarrow dy = dx$

$a = \sqrt{q - \frac{p^2}{4}} = \sqrt{q} = \sqrt{1} = 1$

тогда

$$(x^2+1)^4 = (y^2+1^2)^4$$

$$\int \frac{dy}{(y^2+a^2)^n} = \frac{1}{2(n-1)a^2} \cdot \frac{y}{(y^2+a^2)^{n-1}} + \frac{1}{a^2} \cdot \frac{2 \cdot n - 3}{2 \cdot n - 2} \cdot \int \frac{dy}{(y^2+a^2)^{n-1}}.$$

$$\int \frac{dx}{(x^2+1^2)^4} = \frac{1}{6} \cdot \frac{x}{(x^2+1)^3} + \frac{5}{6} \cdot \int \frac{dx}{(x^2+1)^3} \Bigg|$$

$$\int\frac{dx}{(x^2+1)^3}=\frac{1}{4}\cdot\frac{x}{(x^2+1)^2}+\frac{3}{4}\int\frac{dx}{(x^2+1)^2}$$

$$\int\frac{dx}{(x^2+1)^2}=\frac{1}{2}\cdot\frac{x}{x^2+1}+\frac{1}{2}\int\frac{dx}{x^2+1}$$

$$\int\frac{dx}{x^2+1}=\operatorname{arctg}x+C$$

$\Rightarrow$

$$\Rightarrow\int\frac{dx}{(x^2+1)^4}=\frac{1}{6}\cdot\frac{x}{(x^2+1)^3}+\frac{5}{6}\left(\frac{1}{4}\cdot\frac{x}{(x^2+1)^2}+\frac{3}{4}\left(\frac{1}{2}\cdot\frac{x}{x^2+1}+\frac{1}{2}\cdot\operatorname{arctg}x\right)\right)+$$

$$+C=\frac{x}{6\cdot(x^2+1)^3}+\frac{5x}{24(x^2+1)^2}+\frac{5x}{16(x^2+1)}+\frac{5}{16}\operatorname{arctg}x+C=$$

$$=\frac{8x}{48(x^2+1)^3}+\frac{10x(x^2+1)}{48(x^2+1)^3}+\frac{15x(x^2+1)^2}{48(x^2+1)^3}+\frac{5}{16}\operatorname{arctg}x+C=$$

$$=\frac{8x+10x^3+10x+15x^5+30x^3+15x}{48(x^2+1)^3}+\frac{5}{16}\operatorname{arctg}x+C=$$

$$=\frac{15x^5+40x^3+33x}{48(x^2+1)^3}+\frac{5}{16}\operatorname{arctg}x+C$$

8.3.32

$$\int\frac{(3x+2)dx}{(x^2-3x+3)^2}=\left[\begin{array}{l}x^2-3x+3=0\\ D=9-12=-3<0\Rightarrow\text{нет корней}\\ x^2+px+q=x^2-3x+3\ \text{- во 2-ой степени}\end{array}\right]\Rightarrow\text{интеграл IV типа}$$

$[A=3,\quad B=2,\quad p=-3,\quad q=3$

$$Ax+B=\frac{3}{2}(2x+(-3))+\left(2-\frac{3\cdot(-3)}{2}\right)=\frac{3}{2}(2x-3)+\left(2+\frac{9}{2}\right)$$

$(x^2-3x+3)'_x=2x-3$ - есть в числ. $\Rightarrow$ нах. реш. 1-го инт.

$$t=x^2-3x+3\Rightarrow dt=(2x-3)dx$$

$$\int\frac{(2x-3)dx}{(x^2-3x+3)^2}=\int\frac{dt}{t^2}=\frac{t^{-1}}{-1}+C=-\frac{1}{t}+C=-\frac{1}{x^2-3x+3}+C$$

реш. 2-го инт.

$$y = x + P/2 = x - 3/2 \Rightarrow dy = dx$$

$$a = \sqrt{q - \frac{p^2}{4}} = \sqrt{3 - \frac{9}{4}} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

тогда

$$(x^2 - 3x + 3)^2 = \left(y^2 + \left(\frac{\sqrt{3}}{2}\right)^2\right)^2$$

$$\int \frac{dy}{(y^2+a^2)^n} = \frac{1}{2(n-1)a^2} \cdot \frac{y}{(y^2+a^2)^{n-1}} + \frac{1}{a^2} \cdot \frac{2n-3}{2n-2} \cdot \int \frac{dy}{(y^2+a^2)^{n-1}}$$

$$\int \frac{dy}{(y^2+\frac{3}{4})^2} = \frac{4}{2 \cdot 1 \cdot 3} \cdot \frac{y}{(y^2+\frac{3}{4})^1} + \frac{4}{3} \cdot \frac{1}{2} \cdot \int \frac{dy}{(y^2+\frac{3}{4})} =$$

$$= \left[\int \frac{dx}{x^2+a^2} = \frac{1}{a} \operatorname{arctg} \frac{x}{a} + C\right] = \frac{2y}{3(y^2+\frac{3}{4})} + \frac{2}{3} \cdot \frac{2}{\sqrt{3}} \operatorname{arctg} \frac{2y}{\sqrt{3}} + C =$$

$$= \frac{2y}{3(y^2+\frac{3}{4})} + \frac{4\sqrt{3}}{9} \operatorname{arctg} \frac{2y\sqrt{3}}{3} + C =$$

$$= \frac{2(x-3/2)}{3(x^2-3x+3)} + \frac{4\sqrt{3}}{9} \operatorname{arctg} \frac{2\sqrt{3}(x-3/2)}{3} + C =$$

$$= \frac{2x-3}{3(x^2-3x+3)} + \frac{4\sqrt{3}}{9} \operatorname{arctg} \frac{(2x-3)\sqrt{3}}{3} + C$$

$$\int \frac{Ax+B}{(x^2+px+q)^n} dx = \frac{A}{2} \int \frac{dt}{t^n} + \left(B - \frac{Ap}{2}\right) \cdot \int \frac{dy}{(y^2+a^2)^n} \Bigg] =$$

$$= \frac{3}{2} \cdot \left(-\frac{1}{x^2-3x+3}\right) + \frac{13}{2} \cdot \left(\frac{2x-3}{3(x^2-3x+3)} + \frac{4\sqrt{3}}{9} \operatorname{arctg} \frac{(2x-3)\sqrt{3}}{3}\right) + C =$$

$$= -\frac{3}{2(x^2-3x+3)} + \frac{26x-39}{6(x^2-3x+3)} + \frac{26\sqrt{3}}{9} \operatorname{arctg} \frac{(2x-3)\sqrt{3}}{3} + C =$$

$$= -\frac{9}{6(x^2-3x+3)} + \frac{26x-39}{6(x^2-3x+3)} + \frac{26\sqrt{3}}{9}\,\mathrm{arctg}\,\frac{(2x-3)\sqrt{3}}{3} + C =$$

$$= \frac{26x-48}{6(x^2-3x+3)} + \frac{26\sqrt{3}}{9}\,\mathrm{arctg}\,\frac{(2x-3)\sqrt{3}}{3} + C =$$

$$= \frac{13x-24}{3(x^2-3x+3)} + \frac{26\sqrt{3}}{9}\,\mathrm{arctg}\,\frac{(2x-3)\sqrt{3}}{3} + C$$

8.3.33.

$$\int \frac{2x-3}{(x-1)(x+2)}\,dx = \left[\ \frac{2x-3}{(x-1)(x+2)} = \frac{A}{x-1} + \frac{B}{x+2} = \right.$$

$$= \frac{A(x+2)+B(x-1)}{(x-1)(x+2)}$$

$$\Rightarrow 2x-3 = A(x+2) + B(x-1)$$

$$2x-3 = Ax + 2A + Bx - B$$

$$2x-3 = (Ax+Bx) - (B-2A)$$

$$2x-3 = (A+B)x - (B-2A)$$

$$\begin{cases} A+B=2 \\ B-2A=3 \end{cases} \quad \begin{cases} A+(3+2A)=2 \\ B=3+2A \end{cases} \quad \begin{cases} 3A=-1 \\ B=3+2A \end{cases} \quad \left.\begin{cases} A=-1/3 \\ B=7/3 \end{cases}\ \right] =$$

$$= \int\left(\frac{-\frac{1}{3}}{x-1} + \frac{\frac{7}{3}}{x+2}\right)dx = \int \frac{-\frac{1}{3}}{x-1}\,dx + \int \frac{\frac{7}{3}}{x+2}\,dx = -\frac{1}{3}\int \frac{dx}{x-1} +$$

$$+ \frac{7}{3}\int \frac{dx}{x+2} = \left[\begin{array}{l} 1)\ dx = d(x-1) \\ 2)\ dx = d(x+2) \end{array}\right] = -\frac{1}{3}\int \frac{d(x-1)}{x-1} + \frac{7}{3}\int \frac{d(x+2)}{x+2} =$$

$$= \left[\int \frac{dx}{x} = \ln|x| + C\right] = -\frac{1}{3}\cdot \ln|x-1| + \frac{7}{3}\cdot \ln|x+2| + C$$

8.3.34

$$\int \frac{x-4}{(x-2)(x-3)}\,dx = \left[\frac{x-4}{(x-2)(x-3)} = \frac{A}{x-2} + \frac{B}{x-3} = \right.$$

$$= \frac{A(x-3)+B(x-2)}{(x-2)(x-3)} \Rightarrow x-4 = A(x-3)+B(x-2)$$

$$x-4 = Ax-3A+Bx-2B$$

$$x-4 = (Ax+Bx)-(3A+2B)$$

$$x-4 = (A+B)x-(3A+2B)$$

$$\Rightarrow \begin{cases} A+B=1 \\ 3A+2B=4 \end{cases} \quad \begin{cases} A=1-B \\ 2B=4-3(1-B) \end{cases} \quad \begin{cases} A=1-B \\ 2B-3B=4-3 \end{cases}$$

$$\left. \begin{cases} A=2 \\ B=-1 \end{cases} \right] = \int \left(\frac{2}{x-2} - \frac{1}{x-3} \right) dx = \int \frac{2}{x-2}\,dx - \int \frac{dx}{x-3} =$$

$$= 2\int \frac{dx}{x-2} - \int \frac{dx}{x-3} = \begin{bmatrix} {}^{1)}\, d(x-2)=dx \\ {}^{2)}\, d(x-3)=dx \end{bmatrix} = 2\int \frac{d(x-2)}{x-2} - \int \frac{d(x-3)}{x-3} =$$

$$= \left[\int \frac{dx}{x} = \ln|x| + C \right] = 2\ln|x-2| - \ln|x-3|$$

8.3.35

$$\int \frac{x\,dx}{x^2-4x-5} = \left[\begin{array}{l} x^2-4x-5=0 \\ D=16+20=36 \\ x_{1,2}=\frac{4\pm 6}{2} = \begin{bmatrix} 5 \\ -1 \end{bmatrix} \end{array} \right] = \int \frac{x\,dx}{(x-5)(x+1)} =$$

$$= \left[\frac{x}{(x-5)(x+1)} = \frac{A}{x-5} + \frac{B}{x+1} = \frac{A(x+1)+B(x-5)}{(x-5)(x+1)} \Rightarrow \right.$$

$$\Rightarrow x = A(x+1)+B(x-5)$$

$$x = Ax+A+Bx-5B$$

$$x = (Ax+Bx)+(A-5B)$$

$$x = (A+B)x + (A-5B)$$

$$\Rightarrow \begin{cases} A+B=1 \\ A-5B=0 \end{cases} \quad \begin{cases} B=1-A \\ A-5(1-A)=0 \end{cases} \quad \begin{cases} B=1-A \\ A-5+5A=0 \end{cases} \quad \begin{cases} B=1-A \\ 6A=5 \end{cases}$$

$$\begin{cases} B = 1/6 \\ A = 5/6 \end{cases} \Big] = \int \left(\frac{5/6}{x-5} + \frac{1/6}{x+1} \right) dx =$$

$$= \int \frac{5/6}{x-5} dx + \int \frac{1/6}{x+1} dx = \frac{5}{6} \int \frac{dx}{x-5} + \frac{1}{6} \int \frac{dx}{x+1} =$$

$$= \left[\begin{matrix} 1) \ d(x-5) = dx \\ 2) \ d(x+1) = dx \end{matrix} \right] = \frac{5}{6} \int \frac{d(x-5)}{x-5} + \frac{1}{6} \int \frac{d(x+1)}{x+1} = \left[\int \frac{dx}{x} = \ln|x| + C \right] =$$

$$= \frac{5}{6} \ln|x-5| + \frac{1}{6} \ln|x+1| + C$$

8.3.36.

$$\int \frac{2x^2-11}{x^2+x-6} dx = \left[\begin{array}{r|l} 2x^2+0x-11 & x^2+x-6 \\ 2x^2+2x-12 & 2 \\ \hline -2x+1 & \end{array} \right] =$$

$$= \int 2 dx + \int \frac{-2x+1}{x^2+x-6} dx = 2\int dx - \int \frac{2x-1}{x^2+x-6} dx =$$

$$= \left[\begin{matrix} x^2+x-6=0 \\ D = 1+24 = 25 \\ x_{1,2} = \frac{-1 \pm 5}{2} = \left[\begin{matrix} 2 \\ -3 \end{matrix} \right. \end{matrix} \right] = 2\int dx - \int \frac{2x-1}{(x-2)(x+3)} dx =$$

$$= \left[\frac{2x-1}{(x-2)(x+3)} = \frac{A}{(x-2)} + \frac{B}{(x+3)} = \frac{A(x+3)+B(x-2)}{(x-2)(x+3)} \Rightarrow \right.$$

$$\Rightarrow 2x-1 = A(x+3) + B(x-2)$$

$$2x-1 = Ax + 3A + Bx - 2B$$

$$2x-1 = (Ax+Bx) - (2B-3A)$$

$$2x-1 = (A+B)x - (2B-3A)$$

$$\Rightarrow \begin{cases} A+B=2 \\ 2B-3A=1 \end{cases} \qquad \begin{cases} A=2-B \\ 2B-3(2-B)=1 \end{cases} \qquad \begin{cases} A=2-B \\ 2B+3B=1+6 \end{cases}$$

$$\begin{cases} A=3/5 \\ B=7/5 \end{cases} \Bigg] = 2\int dx - \int \left(\frac{3/5}{(x-2)} + \frac{7/5}{(x+3)} \right) dx =$$

$$= 2\int dx - \frac{3}{5}\int \frac{dx}{x-2} - \frac{7}{5}\int \frac{dx}{x+3} = \left[\begin{array}{l} 1)\ \int dx = x + C \\ 2,3)\ \int \frac{dx}{x} = \ln|x| + C, \ \text{где} \ {}^{2)} d(x-2)=dx, \ {}^{3)} d(x+3)=dx \end{array} \right]$$

$$= 2x - \frac{3}{5}\ln|x-2| - \frac{7}{5}\ln|x+3| + C$$

8.3.37

$$\int \frac{-3x^2+x+19}{(x-4)(x-2)(x+1)}dx = \left[\frac{-3x^2+x+19}{(x-4)(x-2)(x+1)} = \frac{A}{x-4} + \frac{B}{x-2} + \frac{C}{x+1} = \right.$$

$$= \frac{A(x-2)(x+1) + B(x-4)(x+1) + C(x-4)(x-2)}{(x-4)(x-2)(x+1)} \Rightarrow$$

$$\Rightarrow -3x^2+x+19 = A(x-2)(x+1) + B(x-4)(x+1) + C(x-4)(x-2)$$
$$-3x^2+x+19 = A(x^2-2x+x-2) + B(x^2-4x+x-4) + C(x^2-4x-2x+8)$$
$$-3x^2+x+19 = Ax^2 - Ax - 2A + Bx^2 - 3Bx - 4B + Cx^2 - 6Cx + 8C$$
$$-3x^2+x+19 = (Ax^2+Bx^2+Cx^2) + (-Ax-3Bx-6Cx) + (-2A-4B+8C)$$
$$-3x^2+x+19 = (A+B+C)x^2 + (-A-3B-6C)x + (-2A-4B+8C)$$

$$\Rightarrow \begin{cases} A+B+C=-3 \\ A+3B+6C=-1 \\ 2A+4B-8C=-19 \end{cases} \qquad D = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & 6 \\ 2 & 4 & -8 \end{vmatrix} = -24+12+4-6-24+8 = -30 \neq 0$$

$$D_1 = \begin{vmatrix} -3 & 1 & 1 \\ -1 & 3 & 6 \\ -19 & 4 & -8 \end{vmatrix} = 72 - 114 - 4 + 57 + 72 - 8 = 75$$

$$A = \frac{D_1}{D} = -\frac{75}{30} = -\frac{15}{6} = -\frac{5}{2}$$

$$D_2 = \begin{vmatrix} 1 & -3 & 1 \\ 1 & -1 & 6 \\ 2 & -19 & -8 \end{vmatrix} = 8 - 36 - 19 + 2 + 114 - 24 = 45$$

$$B = \frac{D_2}{D} = -\frac{45}{30} = -\frac{9}{6} = -\frac{3}{2}$$

$$D_3 = \begin{vmatrix} 1 & 1 & -3 \\ 1 & 3 & -1 \\ 2 & 4 & -19 \end{vmatrix} = -57 - 2 - 12 + 18 + 4 + 19 = -30$$

$$C = \frac{D_3}{D} = \frac{-30}{-30} = 1$$

$$(A; B; C) = \left(-\frac{5}{2}; -\frac{3}{2}; 1\right) \Big] =$$

$$= \int \frac{-5/2}{x-4}\,dx + \int \frac{-3/2}{x-2}\,dx + \int \frac{dx}{x+1} = -\frac{5}{2}\int \frac{dx}{x-4} - \frac{3}{2}\int \frac{dx}{x-2} + \int \frac{dx}{x+1} =$$

$$= \begin{bmatrix} 1)\ d(x-4) = dx \\ 2)\ d(x-2) = dx \\ 3)\ d(x+1) = dx \end{bmatrix} = -\frac{5}{2}\int \frac{d(x-4)}{x-4} - \frac{3}{2}\int \frac{d(x-2)}{x-2} + \int \frac{d(x+1)}{x+1} =$$

$$= \left[\int \frac{dx}{x} = \ln|x| + C\right] = -\frac{5}{2}\ln|x-4| - \frac{3}{2}\ln|x-2| + \ln|x+1| + C.$$

8.3.38.

$$\int \frac{x-1}{(x+1)(x^2-4)}\,dx = \int \frac{x-1}{(x+1)(x-2)(x+2)}\,dx = \left[\frac{x-1}{(x+1)(x-2)(x+2)} =\right.$$

$$= \frac{A}{x+1} + \frac{B}{x-2} + \frac{C}{x+2} = \frac{A(x-2)(x+2) + B(x+1)(x+2) + C(x+1)(x-2)}{(x+1)(x-2)(x+2)} \Rightarrow$$

$$\Rightarrow \quad x - 1 = A(x-2)(x+2) + B(x+1)(x+2) + C(x+1)(x-2)$$

$$x - 1 = A(x^2-4) + B(x^2 + x + 2x + 2) + C(x^2 - 2x + x - 2)$$

$$x - 1 = Ax^2 - 4A + Bx^2 + 3Bx + 2B + Cx^2 - Cx - 2C$$

$$x - 1 = (A + B + C)x^2 + (3B - C)x + (-4A + 2B - 2C)$$

$$\Leftrightarrow \begin{cases} A+B+C=0 \\ 3B-C=1 \\ -4A+2B-2C=-1 \end{cases} \qquad D=\begin{vmatrix} 1 & 1 & 1 \\ 0 & 3 & -1 \\ -4 & 2 & -2 \end{vmatrix} = -6+4-0+12+2-0=12$$

$$D_1=\begin{vmatrix} 0 & 1 & 1 \\ 1 & 3 & -1 \\ -1 & 2 & -2 \end{vmatrix} = 0+2+1+3-0+2=8$$

$$A=\frac{D_1}{D}=\frac{8}{12}=\frac{2}{3}$$

$$D_2=\begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ -4 & -1 & -2 \end{vmatrix} = -2+0-0+4-1+0=1$$

$$B=\frac{D_2}{D}=\frac{1}{12}$$

$$D_3=\begin{vmatrix} 1 & 1 & 0 \\ 0 & 3 & 1 \\ -4 & 2 & -1 \end{vmatrix} = -3+0-4-0-2-0=-9$$

$$C=\frac{D_3}{D}=\frac{-9}{12}=-\frac{3}{4}$$

$$(A;B;C)=\left(\frac{2}{3};\frac{1}{12};-\frac{3}{4}\right) \quad \Big] =$$

$$=\int\frac{2/3}{x+1}dx+\int\frac{1/12}{x-2}dx+\int\frac{-3/4}{x+2}dx=\frac{2}{3}\int\frac{dx}{x+1}+\frac{1}{12}\int\frac{dx}{x-2}-\frac{3}{4}\int\frac{dx}{x+2}=$$

$$=\begin{bmatrix} 1)\ d(x+1)=dx \\ 2)\ d(x-2)=dx \\ 3)\ d(x+2)=dx \end{bmatrix}=\frac{2}{3}\int\frac{d(x+1)}{x+1}+\frac{1}{12}\int\frac{d(x-2)}{x-2}-\frac{3}{4}\int\frac{d(x+2)}{x+2}=$$

$$=\left[\int\frac{dx}{x}=\ln|x|+C\right]=\frac{2}{3}\ln|x+1|+\frac{1}{12}\ln|x-2|-\frac{3}{4}\ln|x+2|+C$$

8.3.39 $\int \frac{(x^2+2)dx}{(x-1)(x+1)^2} = \Big[\frac{x^2+2}{(x-1)(x+1)^2} = \frac{A}{(x-1)} + \frac{Bx+C}{(x+1)^2} =$

$= \frac{A}{x-1} + \frac{Bx+C}{x^2+2x+1} = \frac{A(x^2+2x+1)+(Bx+C)(x-1)}{(x-1)(x^2+2x+1)} =$

$= \frac{A(x^2+2x+1)+(Bx+C)(x-1)}{(x-1)(x+1)^2}$

$x^2+2 = Ax^2+2Ax+A+Bx^2+Cx-Bx-C$

$x^2+2 = (Ax^2+Bx^2)+(2Ax+Cx-Bx)+(A-C)$

$x^2+2 = (A+B)x^2+(2A+C-B)x+(A-C)$

$\Rightarrow \begin{cases} A+B=1 \\ 2A+C-B=0 \\ A-C=2 \end{cases}$ $\begin{cases} B=1-A \\ 2A+(A-2)-(1-A)=0 \\ C=A-2 \end{cases}$ $\begin{cases} B=1/4 \\ A=3/4 \\ C=-5/4 \end{cases}$

$2A+A-2-1+A=0$ $\quad C=3/4-2 \quad B=1-3/4$

$4A=3 \quad C=-5/4 \quad B=1/4$

$A=3/4$

$(A;B;C)=(3/4;1/4;-5/4)\Big] =$

$= \int \frac{3/4}{(x-1)}dx + \int \frac{(1/4\cdot x - 5/4)}{(x+1)^2}dx =$

$= \frac{3}{4}\int\frac{dx}{x-1} + \int\frac{1/4x}{(x+1)^2}dx - \int\frac{5/4}{(x+1)^2}dx =$

$= \frac{3}{4}\int\frac{dx}{x-1} + \frac{1}{4}\int\frac{x}{(x+1)^2}dx - \frac{5}{4}\int\frac{dx}{(x+1)^2} =$

$= \Big[1) \int\frac{dx}{x-1} = [d(x-1)=dx] = \ln|x-1|+C$

2) $\int\frac{x}{(x+1)^2}dx = \cancel{} = \int\frac{x+1-1}{(x+1)^2}dx = \Big[t=x+1 \to dt = d(x+1) = (x+1)'_x dx = dx \Big]$

$$= \int \frac{t-1}{t^2}\,dt = \int \frac{dt}{t} - \int \frac{dt}{t^2} = \ln|t| + \frac{1}{t} + C =$$

$$= \ln|x+1| + \frac{1}{x+1} + C.$$

$$3) \int \frac{dx}{(x+1)^2} = \left[\begin{array}{l} d(x+1) = dx \\ \int \frac{dx}{x^2} = -\frac{1}{x} + C \end{array} \right] = -\frac{1}{x+1} + C \Bigg] =$$

$$= \frac{3}{4}\ln|x-1| + \frac{1}{4}\ln|x+1| + \frac{1}{4(x+1)} + \frac{5}{4(x+1)} + C =$$

$$= \frac{3}{4}\ln|x-1| + \frac{1}{4}\ln|x+1| + \frac{6}{4(x+1)} + C =$$

$$= \frac{3}{4}\ln|x-1| + \frac{1}{4}\ln|x+1| + \frac{3}{2(x+1)} + C$$

8.3.40.

$$\int \frac{2x+3}{(x-2)^3}\,dx = \int \left(\frac{2x}{(x-2)^3} + \frac{3}{(x-2)^3} \right) dx = \int \frac{2x}{(x-2)^3}\,dx + \int \frac{3}{(x-2)^3}\,dx =$$

$$= 2\int \frac{x}{(x-2)^3}\,dx + 3\int \frac{dx}{(x-2)^3} =$$

$$= \Bigg[\; 1) \int \frac{x}{(x-2)^3}\,dx = \int \frac{x-2+2}{(x-2)^3}\,dx = \left[\begin{array}{l} t = x-2 \Rightarrow dt = d(x-2) = \\ = (x-2)'_x\,dx = dx \end{array} \right] = \int \frac{t+2}{t^3}\,dt =$$

$$= \int \frac{t}{t^3}\,dt + 2\int \frac{dt}{t^3} = \int t^{-2}\,dt + 2\int t^{-3}\,dt = -\frac{1}{t} - \frac{2}{2t^2} + C =$$

$$= -\frac{1}{t} - \frac{2}{2t^2} + C = -\frac{1}{x-2} - \frac{2}{2(x-2)^2} + C$$

$$2) \int \frac{dx}{(x-2)^3} = [d(x-2) = dx] = \int \frac{d(x-2)}{(x-2)^3} = -\frac{1}{2(x-2)^2} + C \Bigg] =$$

$$= -\frac{4}{2(x-2)^2} - \frac{2}{x-2} - \frac{3}{2(x-2)^2} + C = -\frac{2}{x-2} - \frac{7}{2(x-2)^2} + C$$

8.3.41

$$\int \frac{x^4dx}{(x^2-1)(x+2)} = \int \frac{x^4dx}{x^3+2x^2-x-2} = \left[\begin{array}{l|l} x^4+0x^3+0x^2+0x+0 & x^3+2x^2-x-2 \\ \underline{x^4+2x^3-x^2-2x} & x-2 \\ \quad -2x^3+x^2+2x+0 & \\ \quad \underline{-2x^3-4x^2+2x+4} & \\ \qquad 5x^2+0x-4 & \end{array} \right] =$$

$$= \int \frac{(x-2)(x^3+2x^2-x-2)+5x^2-4}{x^3+2x^2-x-2}\,dx = \int (x-2)dx + \int \frac{5x^2-4}{(x^2-1)(x+2)}dx =$$

$$= \left[\frac{5x^2-4}{(x^2-1)(x+2)} = \frac{Ax+B}{x^2-1} + \frac{C}{x+2} = \frac{(Ax+B)(x+2)+C(x^2-1)}{(x^2-1)(x+2)} \right.$$

$$5x^2-4 = Ax^2+2Ax+Bx+2B+Cx^2-C$$

$$5x^2-4 = (A+C)x^2+(2A+B)x+(2B-C)$$

$$\Rightarrow \begin{cases} A+C=5 \\ 2A+B=0 \\ 2B-C=-4 \end{cases} \qquad \begin{cases} A=5-C \\ 2(5-C)+(-2+\frac{1}{2}C)=0 \\ B=-2+\frac{1}{2}C \end{cases} \qquad \begin{array}{l} 10-2C-2+\frac{1}{2}C=0 \\ \frac{3}{2}C=8 \\ C=16/3 \end{array}$$

$$\begin{cases} A=-1/3 \\ C=16/3 \\ B=2/3 \end{cases} \qquad \left.(A;B;C)=\left(-\frac{1}{3};\frac{2}{3};\frac{16}{3}\right)\right] =$$

$$= \int (x-2)dx + \int \frac{-\frac{1}{3}x+\frac{2}{3}}{x^2-1}dx + \int \frac{\frac{16}{3}}{x+2}dx = \int (x-2)dx - \frac{1}{3}\int \frac{xdx}{x^2-1} + \frac{2}{3}\int \frac{dx}{x^2-1} + \frac{16}{3}\int \frac{dx}{x+2} =$$

$$= \left[1)\int \frac{xdx}{x^2-1} = \left[\begin{array}{l} t=x^2-1 \Rightarrow dt=d(x^2-1)= \\ =(x^2-1)'_x dx = 2xdx \end{array} \right] = \int \frac{dt}{2t} = \frac{1}{2}\int \frac{dt}{t} = \frac{1}{2}\ln|t|+C = \right.$$

$$= \frac{1}{2}\ln|x^2-1|+C.$$

$$2)\int \frac{dx}{x^2-1} = \left[\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\ln\left|\frac{x-a}{x+a}\right| + C \right] = \frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + C$$

$$3)\int \frac{dx}{x+2} = [d(x+2)=dx] = \int \frac{d(x+2)}{x+2} = \ln|x+2|+C$$

$$\left. 1)\int (x-2)dx = \int xdx - \int 2dx = \frac{x^2}{2} - 2x + C \right] =$$

$$= \frac{x^2}{2} - 2x - \frac{1}{3}\cdot\frac{1}{2}\cdot \ln|x^2-1| + \frac{2}{3}\cdot\frac{1}{2}\ln\left|\frac{x-1}{x+1}\right| + \frac{16}{3}\ln|x+2| + C =$$

$$= \frac{3x^2}{6} - \frac{12x}{6} - \frac{1}{6}\ln|x^2-1| + \frac{2}{6}\ln\left|\frac{x-1}{x+1}\right| + \frac{32}{6}\ln|x+2| + C =$$

$$= \frac{1}{6}\left(3x^2 - 12x - \ln|x^2-1| + 2\ln|x-1| - 2\ln|x+1| + 32\ln|x+2|\right) + C =$$

$$= \frac{1}{6}\left(3x^2 - 12x + \ln\left|\frac{(x-1)^2}{(x-1)(x+1)}\right| - 2\ln|x+1| + 32\ln|x+2|\right) + C =$$

$$= \frac{1}{6}\left(3x^2 - 12x + \ln\left|\frac{x-1}{x+1}\right| - 2\ln|x+1| + 32\ln|x+2|\right) + C =$$

$$= \frac{1}{6}\left(3x^2 - 12x + \ln|x-1| - \ln|x+1| - 2\ln|x+1| + 32\ln|x+2|\right) + C =$$

$$= \frac{1}{6}\left(3x^2 - 12x + \ln|x-1| - 3\ln|x+1| + 32\ln|x+2|\right) + C$$

8.3.42

$$\int \frac{dx}{x^3-1} = \int \frac{dx}{(x-1)(x^2+x+1)} = \left[\frac{1}{(x-1)(x^2+x+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+x+1} =\right.$$

$$= \frac{A(x^2+x+1) + (Bx+C)(x-1)}{(x-1)(x^2+x+1)} \Rightarrow \begin{aligned} &1 = A(x^2+x+1) + (Bx+C)(x-1) \\ &1 = Ax^2 + Ax + A + Bx^2 - Bx + Cx - C \\ &1 = (A+B)x^2 + (A-B+C)x + (A-C) \end{aligned}$$

$$\Rightarrow \begin{cases} A+B=0 \\ A-B+C=0 \\ A-C=1 \end{cases} \quad \begin{cases} B=-A \\ A-(-A)+(A-1)=0 \\ C=A-1 \end{cases} \quad \begin{cases} B=-1/3 \\ A=1/3 \\ C=-2/3 \end{cases}$$

$$\left.(A;B;C) = (1/3; -1/3; -2/3)\right] =$$

$$= \int \frac{1/3}{x-1}dx + \int \frac{-1/3\cdot x - 2/3}{x^2+x+1}dx = \frac{1}{3}\int\frac{dx}{x-1} - \int\frac{\frac{1}{3}x+\frac{2}{3}}{x^2+x+1} =$$

$$= \Big[\ ^{1)} \int\frac{dx}{x-1} = [d(x-1)=dx] = \int\frac{d(x-1)}{x-1} = \ln|x-1| + C$$

$$2)\ \int \frac{\frac{1}{3}x+\frac{2}{3}}{x^2+x+1} = \left[\text{III тип: } A=\frac{1}{3};\ B=\frac{2}{3};\ p=1;\ q=1\right] =$$

$$= \frac{1}{6}\int \frac{2x+1}{x^2+x+1}\,dx + \left(\frac{2}{3}-\frac{1}{6}\right)\int \frac{dx}{x^2+x+} =$$

$$= \frac{1}{6}\ln|x^2+x+1| + \frac{3}{6}\cdot\frac{2}{\sqrt{4-1^2}}\,\mathrm{arctg}\,\frac{2x+1}{\sqrt{4-1^2}} + C =$$

$$= \frac{1}{6}\ln|x^2+x+1| + \frac{1}{\sqrt{3}}\,\mathrm{arctg}\,\frac{2x+1}{\sqrt{3}} + C\ \Big] =$$

$$= \frac{1}{3}\cdot\ln|x-1| - \frac{1}{6}\ln(x^2+x+1) - \frac{1}{\sqrt{3}}\,\mathrm{arctg}\,\frac{2x+1}{\sqrt{3}} + C$$

8.3.43

$$\int \frac{x\,dx}{(x^2-1)(x^2+1)} = \left[\frac{x}{(x^2-1)(x^2+1)} = \frac{Ax+B}{(x^2-1)} + \frac{Cx+D}{(x^2+1)} = \right.$$

$$= \frac{(Ax+B)(x^2+1) + (Cx+D)(x^2-1)}{(x^2-1)(x^2+1)}$$

$$\Rightarrow x = Ax^3 + Bx^2 + Ax + B + Cx^3 + Dx^2 - Cx - D$$

$$x = (A+C)x^3 + (B+D)x^2 + (A-C)x + (B-D)$$

$$\Rightarrow \begin{cases} A+C=0 \\ B+D=0 \\ A-C=1 \\ B-D=0 \end{cases} \qquad \left.\begin{matrix} A = 1+C \\ A = -C \end{matrix}\right| \Rightarrow 1+C = -C$$

$$2C = -1$$

$$C = -1/2 \quad \Rightarrow A = 1/2$$

$$\left.\begin{matrix} B = -D \\ B = D \end{matrix}\right| \Rightarrow -D = D \Rightarrow D = 0 \Rightarrow B = 0$$

$$\left.(A;B;C;D) = (1/2; 0; -1/2; 0)\right] =$$

$$= \int \frac{1/2 \cdot x}{x^2-1}\,dx + \int \frac{-1/2 \cdot x}{x^2+1}\,dx = \frac{1}{2}\int \frac{x\,dx}{x^2-1} - \frac{1}{2}\int \frac{x\,dx}{x^2+1} =$$

$$= \left[\begin{matrix} 1)\ t = x^2-1 \Rightarrow dt = d(x^2-1) = (x^2-1)'_x\,dx = 2x\,dx \\ 2)\ y = x^2+1 \Rightarrow dy = d(x^2+1) = (x^2+1)'_x\,dx = 2x\,dx \end{matrix} \right] =$$

$$= \frac{1}{4}\int \frac{dt}{t} - \frac{1}{4}\int \frac{dy}{y} = \frac{1}{4}\ln|x^2-1| - \frac{1}{4}\ln|x^2+1| =$$

$$= \frac{1}{4}\ln|x^2-1| - \frac{1}{4}\ln(x^2+1)$$

8.3.44

$$\int \frac{dx}{(x^2+1)(x^2+4)} = \left[\frac{1}{(x^2+1)(x^2+4)} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{x^2+4} = \right.$$

$$= \frac{(Ax+B)(x^2+4) + (Cx+D)(x^2+1)}{(x^2+1)(x^2+4)}$$

$$\Rightarrow 1 = Ax^3 + Bx^2 + 4Ax + 4B + Cx^3 + Dx^2 + Cx + D$$
$$1 = (A+C)x^3 + (B+D)x^2 + (4A+C)x + (4B+D)$$

$$\Rightarrow \begin{cases} A+C=0 \\ B+D=0 \\ 4A+C=0 \\ 4B+D=1 \end{cases} \qquad \begin{matrix} D = 1-4B \\ D=-B \end{matrix} \Big| \Rightarrow \begin{matrix} 1-4B=-B \\ B = 1/3 \end{matrix} \Rightarrow D = -1/3$$

$$\begin{matrix} C=-4A \\ C=-A \end{matrix} \Big| \Rightarrow \begin{matrix} -4A=-A \\ A=0 \end{matrix} \Rightarrow C=0$$

$$\left. (A;B;C;D) = (0; 1/3; 0; -1/3) \right] =$$

$$= \int \frac{1/3}{x^2+1}dx + \int \frac{-1/3}{x^2+4}dx = \frac{1}{3}\int \frac{dx}{x^2+1} - \frac{1}{3}\int \frac{dx}{x^2+4} =$$

$$= \frac{1}{3}\operatorname{arctg} x - \frac{1}{3}\cdot\frac{1}{2}\operatorname{arctg}\frac{x}{2} + C =$$

$$= \frac{1}{3}\operatorname{arctg} x - \frac{1}{6}\operatorname{arctg}\frac{x}{2} + C$$

8.3.45

$$\int \frac{2x^2-3x-3}{(x^2-2x+5)(x-1)}dx = \left[\frac{2x^2-3x-3}{(x^2-2x+5)(x-1)} = \frac{Ax+B}{x^2-2x+5} + \frac{C}{x-1} = \right.$$

$$= \frac{(Ax+B)(x-1) + C(x^2-2x+5)}{(x^2-2x+5)(x-1)} \qquad \Rightarrow 2x^2-3x-3 = Ax^2+Bx-Ax-B+Cx^2-2Cx+5C$$
$$2x^2-3x-3 = (A+C)x^2 + (B-A-2C)x + (5C-B)$$

$$\Rightarrow \begin{cases} A+C=2 \\ B-A-2C=-3 \\ 5C-B=-3 \end{cases} \qquad \begin{cases} A=2-C \\ 5C+3-2+C-2C=-3 \\ B=5C+3 \end{cases} \qquad \begin{cases} A=3 \\ C=-1 \\ B=-2 \end{cases}$$

$$(A;B;C) = (3;-2;-1) \,]=$$

$$= \int \frac{3x-2}{x^2-2x+5}\,dx - \int \frac{dx}{x-1} = \Big[1) \int \frac{3x-2}{x^2-2x+5}\,dx = [\text{III тип}: A=3;\ B=-2;\ p=-2;\ q=5\ ] =$$

$$= \frac{3}{2}\cdot \ln(x^2-2x+5) + \frac{2\cdot(-2)-3\cdot(-2)}{\sqrt{4\cdot5-(-2)^2}}\cdot \operatorname{arctg}\frac{2x-2}{\sqrt{4\cdot5-(-2)^2}} + C =$$

$$= \frac{3}{2}\cdot \ln(x^2-2x+5) + \frac{2}{4}\operatorname{arctg}\frac{2x-2}{4} + C =$$

$$= \frac{3}{2}\ln(x^2-2x+5) + \frac{1}{2}\operatorname{arctg}\frac{x-1}{2} + C$$

$$2) \int \frac{dx}{x-1} = [d(x-1)=dx] = \int \frac{d(x-1)}{x-1} = \ln|x-1| + C \,\Big] =$$

$$= \frac{3}{2}\ln(x^2-2x+5) + \frac{1}{2}\operatorname{arctg}\frac{x-1}{2} + \ln|x-1| + C$$

8.5.46

$$\int \frac{x^4+x^3+x^2+x+1}{(x^2+1)^2\cdot x}\,dx = \Big[\frac{x^4+x^3+x^2+x+1}{(x^2+1)^2\cdot x} = \frac{Ax+B}{x^2+1} + \frac{Cx+D}{(x^2+1)^2} + \frac{E}{x} =$$

$$= \frac{(Ax+B)(x^2+1)\cdot x + (Cx+D)\,x + E(x^2+1)^2}{(x^2+1)^2\cdot x} \Rightarrow$$

$$\Rightarrow x^4+x^3+x^2+x+1 = (Ax+B)(x^2+1)x + (Cx+D)x + E(x^2+1)^2$$

$$x^4+x^3+x^2+x+1 = (Ax+B)(x^3+x) + Cx^2+Dx + E(x^4+2x^2+1)$$

$$x^4+x^3+x^2+x+1 = Ax^4+Bx^3+Ax^2+Bx+Cx^2+Dx+Ex^4+2Ex^2+E$$

$$x^4+x^3+x^2+x+1 = (A+E)x^4 + (B)x^3 + (A+C+2E)x^2 + (B+D)x + E$$

$$\Rightarrow \begin{cases} A+E=1 \\ B=1 \\ A+C+2E=1 \\ B+D=1 \\ E=1 \end{cases} \qquad \begin{cases} A=0 \\ B=1 \\ C=-1 \\ D=0 \\ E=1 \end{cases} \qquad (A;B;C;D;E) = (0;1;-1;0;1) \Big] =$$

$$= \int \frac{dx}{x^2+1} - \int \frac{x\,dx}{(x^2+1)^2} + \int \frac{dx}{x} =$$

$$= \Big[\; 1) \int \frac{dx}{x^2+1} = \operatorname{arctg} x + C$$

$$2) \int \frac{x\,dx}{(x^2+1)^2} = \left[\begin{array}{l} t = x^2+1 \Rightarrow dt = d(x^2+1) = (x^2+1)'_x\,dx = \\ = 2x\,dx \Rightarrow dt/2 = x\,dx \end{array} \right] =$$

$$= \frac{1}{2} \int \frac{dt}{t^2} = \frac{1}{2} \cdot \left(-\frac{1}{t}\right) + C = -\frac{1}{2(x^2+1)} + C$$

$$3) \int \frac{dx}{x} = \ln|x| + C \quad \Big] =$$

$$= \operatorname{arctg} x + \frac{1}{2(x^2+1)} + \ln|x| + C$$

8.3.47.

$$\int \frac{x^4 - 2x^3 + 3x + 4}{x^3+1}\,dx = \left[\begin{array}{l|l} x^4 - 2x^3 + 0x^2 + 3x + 4 & x^3+1 \\ \underline{x^4 + 0x^3 + 0x^2 + x} & x - 2 \\ \quad -2x^3 \qquad\quad + 2x & \\ \quad \underline{-2x^3 \qquad\quad + 0x - 2} & \\ \qquad\qquad\qquad 2x + 6 & \end{array} \right] =$$

$$= \int \frac{(x^2+1)(x-2) + 2x + 6}{x^3+1}\,dx = \int (x-2)\,dx + \int \frac{2x+6}{x^3+1}\,dx =$$

$$= \Big[\; 1) \int (x-2)\,dx = [d(x-2) = dx] = \frac{(x-2)^2}{2} + C = \frac{x^2}{2} - 2x + C$$

$$2) \int \frac{2x+6}{x^3+1}\,dx = \Big| \frac{2x+6}{x^3+1} = \frac{2x+6}{(x+1)(x^2-x+1)} = \frac{A}{x+1} + \frac{Bx+C}{x^2-x+1} =$$

$$= \frac{A(x^2-x+1) + (Bx+C)(x+1)}{(x+1)(x^2-x+1)} \Rightarrow 2x+6 = Ax^2 - Ax + A + Bx^2 + Cx + Bx + C$$

$$2x + 6 = (A+B)x^2 + (-A+C+B)x + (A+C)$$

$$\Rightarrow \begin{cases} A+B=0 \\ -A+B+C=2 \\ A+C=6 \end{cases} \qquad \begin{cases} A=-B \\ B+B+6+B=2 \\ C=6-A \end{cases} \qquad \begin{cases} A=4/3 \\ B=-4/3 \\ C=14/3 \end{cases}$$

$$(A;B;C)=\left(\frac{4}{3};\ -\frac{4}{3};\ \frac{14}{3}\right)\Big]^{3} =$$

$$= \frac{4}{3}\int\frac{dx}{x+1} - \int\frac{\frac{4}{3}x-\frac{14}{3}}{x^2-x+1}dx = \Big[^{4}\ \text{тип III}: A=\frac{4}{3};\ B=-\frac{14}{3};\ p=-1;\ q=1\Big]^{4} =$$

$$\Big[^{5}\int\frac{\frac{4}{3}x-\frac{14}{3}}{x^2-x+1}dx = \frac{2}{3}\cdot\ln(x^2-x+1) + \frac{-\frac{28}{3}+\frac{4}{3}}{\sqrt{4-1}}\cdot\operatorname{arctg}\frac{2x-1}{\sqrt{4-1}} + C =$$

$$= \frac{2}{3}\ln(x^2-x+1) - \frac{24}{3\sqrt{3}}\cdot\operatorname{arctg}\frac{2x-1}{\sqrt{3}} + C =$$

$$= \frac{2}{3}\ln(x^2-x+1) - \frac{8}{\sqrt{3}}\operatorname{arctg}\frac{2x-1}{\sqrt{3}} + C\ \Big]^{5} = \Big[^{6}\ d(x+1)=dx\Big]^{6} =$$

$$= \frac{4}{3}\ln|x+1| - \frac{2}{3}\ln(x^2-x+1) + \frac{8}{\sqrt{3}}\operatorname{arctg}\frac{2x-1}{\sqrt{3}} + C\ \Big]^{1} =$$

$$= \frac{x^2}{2} - 2x + \frac{4}{3}\ln|x+1| - \frac{2}{3}\ln(x^2-x+1) + \frac{8}{\sqrt{3}}\operatorname{arctg}\frac{2x-1}{\sqrt{3}} + C$$

8.3.48

$$\int\frac{3x+5}{x(x^2+1)^2}dx = \Big[^{1}\ \frac{3x+5}{x(x^2+1)^2} = \frac{A}{x} + \frac{Bx+C}{x^2+1} + \frac{Dx+E}{(x^2+1)^2} =$$

$$= \frac{A(x^2+1)^2 + (Bx+C)\cdot(x^2+1)\cdot x + (Dx+E)x}{x(x^2+1)^2} \Rightarrow$$

$$\Rightarrow 3x+5 = Ax^4 + 2Ax^2 + A + Bx^4 + Cx^3 + Bx^2 + Cx + Dx^2 + Ex$$

$$3x+5 = (A+B)x^4 + Cx^3 + (2A+B+D)x^2 + (C+E)x + A$$

$$\Rightarrow \begin{cases} A+B=0 \\ C=0 \\ 2A+B+D=0 \\ C+E=3 \\ A=5 \end{cases} \qquad \begin{cases} B=-5 \\ C=0 \\ D=-5 \\ E=3 \\ A=5 \end{cases}$$

$$(A;B;C;D;E)=(5;-5;0;-5;3)\Big]^1 =$$

$$= 5\int\frac{dx}{x} - 5\int\frac{x\,dx}{x^2+1} - \int\frac{5x-3}{(x^2+1)^2} =$$

$$= \Big[^2 \; 1)\ \int\frac{dx}{x} = \ln|x| + C$$

$$2)\ \int\frac{x\,dx}{x^2+1} = \left[t=x^2+1 \Rightarrow dt=d(x^2+1)=(x^2+1)'_x\,dx=2x\,dx\right] = \frac{1}{2}\int\frac{dt}{t} = \frac{1}{2}\ln|t|+C =$$

$$= \frac{1}{2}\ln|x^2+1|+C.$$

$$3)\ \int\frac{5x-3}{(x^2+1)^2} = \left[\text{IV тип}: A=5;\ B=-3;\ p=0;\ q=1;\ n=2;\ y=x;\ a=1\right] =$$

$$= \frac{5}{2(-1)}\cdot\frac{1}{(x^2+1)} + (-3)\cdot\int\frac{dx}{(x^2+1)^2} =$$

$$= -\frac{5}{2(x^2+1)} - 3\cdot\left(\frac{1}{2}\cdot\frac{x}{x^2+1}+\frac{1}{2}\cdot\operatorname{arctg}x\right)+C\Big]^2 =$$

$$= 5\ln|x| - 5\cdot\frac{1}{2}\ln|x^2+1| + \frac{5}{2(x^2+1)} + \frac{3x}{2(x^2+1)} + \frac{3}{2}\operatorname{arctg}x + C =$$

$$= 5\ln|x| - \frac{5}{2}\ln(x^2+1) + \frac{3x+5}{2(x^2+1)} + \frac{3}{2}\operatorname{arctg}x + C.$$

8.3.49

$$\int\frac{x^2\,dx}{x^3+5x^2+8x+4} = \left[\begin{aligned} &x^3+5x^2+8x+4 = x^3+x^2+4x^2+4x+4x+4 = \\ &= x^3+(x+2)^2+4x^2+4x = x(x^2+4x+4)+(x+2)^2 = (x+2)^2(x+1) \end{aligned}\right] \Rightarrow$$

$$\Rightarrow \frac{x^2}{(x+2)^2(x+1)} = \frac{A}{(x+2)^2} + \frac{B}{x+2} + \frac{C}{x+1} = \frac{A(x+1)+B(x+2)(x+1)+C(x+2)^2}{(x+2)^2(x+1)}$$

$$\Rightarrow x^2 = Ax + A + Bx^2 + 2Bx + Bx + 2B + Cx^2 + 4Cx + 4C$$

$$x^2 = (B+C)x^2 + (A+3B+4C)x + (A+2B+4C)$$

$$\Rightarrow \begin{cases} B+C=1 \\ A+3B+4C=0 \\ A+2B+4C=0 \end{cases} \qquad \begin{cases} C=1-B \\ -2B-4C+3B+4C=0 \\ A=-2B-4C \end{cases} \qquad \begin{cases} C=1 \\ B=0 \\ A=-4 \end{cases}$$

$$(A;B;C) = (-4;0;1)\Big] =$$

$$= \int \frac{-4}{(x+2)^2}dx + \int \frac{0}{x+2}dx + \int \frac{dx}{x+1} = -4\int \frac{dx}{(x+2)^2} + \int \frac{dx}{x+1} =$$

$$= \begin{bmatrix} 1)\ d(x+2)=dx \\ 2)\ d(x+1)=dx \end{bmatrix} = -4\int \frac{d(x+2)}{(x+2)^2} + \int \frac{d(x+1)}{x+1} = \frac{4}{x+2} + \ln|x+1| + C$$

8.3.50

$$\int \frac{dt}{t^4-1} = \int \frac{dt}{(t^2-1)(t^2+1)} = \int \frac{dt}{(t-1)(t+1)(t^2+1)} =$$

$$= \Big[\frac{1}{(t-1)(t+1)(t^2+1)} = \frac{A}{t-1} + \frac{B}{t+1} + \frac{Ct+D}{t^2+1} =$$

$$= \frac{A(t+1)(t^2+1)+B(t-1)(t^2+1)+(Ct+D)(t^2-1)}{(t-1)(t+1)(t^2+1)}$$

$$\Rightarrow 1 = A(t+1)(t^2+1)+B(t-1)(t^2+1)+(Ct+D)(t^2-1)$$

$$1 = At^3+At^2+At+A+Bt^3-Bt^2+Bt-B+Ct^3+Dt^2-Ct-D$$

$$1 = (A+B+C)t^3+(A-B+D)t^2+(A+B-C)t+(A-B-D)$$

$$\Rightarrow \begin{cases} A+B+C=0 \\ A-B+D=0 \\ A+B-C=0 \\ A-B-D=1 \end{cases} \qquad D = \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & -1 & 0 \\ 1 & -1 & 0 & -1 \end{vmatrix} = \begin{vmatrix} -1 & 0 & 1 \\ 1 & -1 & 0 \\ -1 & 0 & -1 \end{vmatrix} - \begin{vmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \\ 1 & 0 & -1 \end{vmatrix} + \begin{vmatrix} 1 & -1 & 1 \\ 1 & 1 & 0 \\ 1 & -1 & 4 \end{vmatrix} =$$

$$= -1-1-1-1+(-1)-1-1-1 = -8$$

$$D_1 = \begin{vmatrix} 0 & 1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 1 & -1 & 0 & -1 \end{vmatrix} = -\begin{vmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{vmatrix} + \begin{vmatrix} 0 & -1 & 1 \\ 0 & 1 & 0 \\ 1 & -1 & -1 \end{vmatrix} = -1-1 = -2$$

$$A = \frac{D_1}{D} = \frac{-2}{-8} = \frac{1}{4}$$

$$D_2 = \begin{vmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & -1 & 0 \\ 1 & 1 & 0 & -1 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 \\ 0 & -1 & 0 \\ 1 & 0 & -1 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & -1 \end{vmatrix} = 1+1 = 2$$

$$B = \frac{D_2}{D} = \frac{2}{-8} = -\frac{1}{4}$$

$$D_3 = \begin{vmatrix} 1 & 1 & 0 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 1 & -1 & 1 & -1 \end{vmatrix} = \begin{vmatrix} -1 & 0 & 1 \\ 1 & 0 & 0 \\ -1 & 1 & -1 \end{vmatrix} - \begin{vmatrix} 1 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & -1 \end{vmatrix} = 1-1 = 0$$

$$C = \frac{D_3}{D} = \frac{0}{-8} = 0$$

$$D_4 = \begin{vmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ 1 & 1 & -1 & 0 \\ 1 & -1 & 0 & 1 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 1 & 1 & -1 \end{vmatrix} = 1+1+1+1 = 4$$

$$D = \frac{D_4}{D} = \frac{4}{-8} = -\frac{1}{2}$$

$$(A; B; C; D) = \left(\frac{1}{4}; -\frac{1}{4}; 0; -\frac{1}{2}\right) \Big] =$$

$$= \frac{1}{4}\int \frac{dt}{t-1} - \frac{1}{4}\int \frac{dt}{t+1} - \frac{1}{2}\int \frac{dt}{t^2+1} =$$

$$= \left[\begin{aligned} &1)\int \frac{dt}{t-1} = [d(t-1)=dt] = \ln|t-1| + C \\ &2)\int \frac{dt}{t+1} = [d(t+1)=dt] = \ln|t+1| + C \\ &3)\int \frac{dt}{t^2+1} = \operatorname{arctg} t + C \end{aligned} \right] =$$

$$= \frac{1}{4}\ln|t-1| - \frac{1}{4}\ln|t+1| - \frac{1}{2}\operatorname{arctg} t + C$$

8.351

$$\int \frac{e^{2x}dx}{e^{2x}+3e^x+2} = \left[t = e^x \Rightarrow dt = d(e^x) = (e^x)'_x\,dx = e^x dx\right] = \int \frac{t^2\,dt}{t(t^2+3t+2)} =$$

$$= \left[\begin{aligned} &t^2+3t+2=0 \quad D = 9-8=1 \\ &x_{1,2} = \frac{-3\pm 1}{2} = \left[\begin{aligned}&-1\\&-2\end{aligned}\right. \end{aligned} \right] = \int \frac{t\,dt}{(t+1)(t+2)} = \left[\frac{t}{(t+1)(t+2)} = \frac{A}{t+1} + \frac{B}{t+2} = \right.$$

$$= \frac{A(t+2)+B(t+1)}{(t+1)(t+2)} \Rightarrow \begin{aligned} t &= At + 2A + Bt + B \\ t &= (A+B)t + (2A+B) \end{aligned}$$

$$\Rightarrow \begin{cases} A+B=1 \\ 2A+B=0 \end{cases} \qquad \begin{aligned} -A &= 1 \\ A &= -1 \end{aligned} \qquad \begin{aligned} &2\cdot(-1)+B=0 \\ &B=2 \end{aligned}$$

$$\begin{cases} A = -1 \\ B = 2 \end{cases} \qquad \left.(A;B) = (-1;2)\right] =$$

$$= -\int \frac{dt}{t+1} + 2\int\frac{dt}{t+2} = \left[\begin{aligned} &1)\ d(t+1) = dt \\ &2)\ d(t+2) = dt \end{aligned}\right] = -\ln|t+1| + 2\ln|t+2| + C =$$

$$= 2\ln(e^x+2) - \ln(e^x+1) + C.$$

8.3.52.

$$\int \frac{1+e^x}{(1-e^{2x})e^x}dx = \left[t=e^x \Rightarrow dt=e^x dx\right] = \int \frac{1+t}{(1-t^2)t^2}dt =$$

$$= \left[\frac{1+t}{t^2(1-t^2)} = \frac{At+B}{t^2} + \frac{Ct+D}{1-t^2} = \frac{(At+B)(1-t^2)+(Ct+D)t^2}{t^2(1-t^2)} \right.$$

$$\Rightarrow 1+t = At - At^3 + B - Bt^2 + Ct^3 + Dt^2$$
$$1+t = (-A+C)t^3 + (-B+D)t^2 + (A)t + B$$

$$\Rightarrow \begin{cases} -A+C=0 \\ -B+D=0 \\ A=1 \\ B=1 \end{cases} \quad \begin{cases} C=1 \\ D=1 \\ A=1 \\ B=1 \end{cases} \quad \left.(A;B;C;D)=(1;1;1;1)\right] =$$

$$= \int \frac{t+1}{t^2}dt + \int \frac{t+1}{1-t^2}dt = \int \frac{dt}{t} + \int \frac{dt}{t^2} + \int \frac{(1+t)dt}{(1-t)(1+t)} =$$

$$= \int \frac{dt}{t} + \int \frac{dt}{t^2} + \int \frac{dt}{1-t} = [d(1-t)=-dt] = \ln|t| - \frac{1}{t} - \ln|1-t| + C =$$

$$= \ln|e^x| - \frac{1}{e^x} - \ln|1-e^x| + C = x - \frac{1}{e^x} - \ln|1-e^x| + C.$$

8.3.53

$$\int \frac{\cos x\,dx}{(\sin x - 1)(\sin x + 2)} = \left[\begin{matrix} t=\sin x \Rightarrow dt = d(\sin x) = \\ =(\sin x)'_x dx = \cos x\,dx \end{matrix}\right] = \int \frac{dt}{(t-1)(t+2)} =$$

$$= \left[\frac{1}{(t-1)(t+2)} = \frac{A}{t-1} + \frac{B}{t+2} = \frac{A(t+2)+B(t-1)}{(t-1)(t+2)} \right.$$

$$\Rightarrow 1 = At + 2A + Bt - B$$
$$1 = (A+B)t + (2A-B)$$

$$\Rightarrow \begin{cases} A+B=0 \\ 2A-B=1 \end{cases} \quad \begin{cases} A=-B \\ -3B=1 \end{cases} \quad \begin{cases} A=1/3 \\ B=-1/3 \end{cases} \quad \left.(A;B)=(1/3;-1/3)\right] =$$

$$= \frac{1}{3}\int\frac{dt}{t-1} - \frac{1}{3}\int\frac{dt}{t+2} = \left[\begin{matrix} 1)\ d(t-1) = dt \\ 2)\ d(t+2) = dt \end{matrix}\right] =$$

$$= \frac{1}{3}\ln|t-1| - \frac{1}{3}\ln|t+2| + C =$$

$$= \frac{1}{3}\ln|\sin x - 1| - \frac{1}{3}\ln|\sin x + 2| + C =$$

$$= \frac{1}{3}\ln|\sin x - 1| - \frac{1}{3}\ln(\sin x + 2) + C.$$

8.3.54.

$$\int\frac{\sin^4 x\,dx}{\cos x} = \int\frac{\cos x\sin^4 x\,dx}{1-\sin^2 x} = \left[t = \sin^2 x \Rightarrow dt = d(\sin^2 x) = (\sin^2 x)'_x\,dx = 2\cos x\,dx\right]$$

$$= \int\frac{t^2}{2(1-t)}dt = \frac{1}{2}\int\frac{t^2}{1-t}dt = \left[\begin{array}{r|l} t^2+0t+0 & -t+1 \\ \underline{t^2-t\quad} & \overline{-t-1} \\ t+0 & \\ \underline{t-1} & \\ 1 & \end{array}\right] =$$

$$= \frac{1}{2}\int\left(\frac{(-t-1)(1-t)}{1-t} + \frac{1}{1-t}\right)dt = \frac{1}{2}\left(\int(t+1)\,dt + \int\frac{dt}{1-t}\right) =$$

$$= -\frac{1}{2}\int(t+1)\,dt + \frac{1}{2}\int\frac{dt}{1-t} = \left[\begin{matrix} 1)\ d(t+1) = dt \\ 2)\ d(1-t) = -dt \end{matrix}\right] =$$

$$= -\frac{1}{2}\cdot\frac{(t+1)^2}{2} - \frac{1}{2}\cdot\ln|1-t| + C = -\frac{(t+1)^2}{4} - \frac{1}{2}\ln|1-t| + C =$$

$$= -\frac{t^2}{4} - \frac{t}{2} - \frac{1}{2}\ln|1-t| + C = -\frac{\sin^4 x}{4} - \frac{\sin^2 x}{2} - \frac{1}{2}\ln|\cos^2 x| + C.$$